Md. Farhad Momin

Lecturer, Department of Civil Engineering World University of Bangladesh

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PROFILE

Lecturer in Civil Engineering and M.Sc. candidate in Structural Engineering at BUET. My work brings together earthquake engineering, structural assessment and data-driven methods, with a focus on the seismic assessment of existing reinforced concrete buildings, sustainable construction materials, and machine learning for predicting structural and material behavior.

APPOINTMENT

Lecturer July 2022 to Present
Department of Civil Engineering, World University of Bangladesh

EDUCATION

M.Sc. in Structural Engineering April 2022 to Present
Bangladesh University of Engineering and Technology (BUET)
Thesis: comparison of seismic assessment methods for reinforced concrete buildings. Supervisor: Prof. Dr. Raquib Ahsan.

B.Sc. in Civil Engineering April 2017 to January 2022
Ahsanullah University of Science and Technology (AUST)
Thesis: seismic vulnerability assessment of an existing RC building. Supervisor: Dr. Debasish Sen.

RESEARCH

Interests

Structural engineering, earthquake engineering, seismic evaluation of reinforced concrete buildings, structural dynamics, sustainable construction materials, and machine-learning applications in civil engineering.

Current and recent work

- Seismic assessment of existing buildings in Bangladesh
- Machine-learning prediction of ultra-high-performance concrete strength
- Data-driven estimation of the fundamental period of steel-braced RC frames

PUBLICATIONS

JOURNAL ARTICLES

1 Rahman, T., **Momin, Md. F.**, Podder, S. K., Provasha, A. A., & Kumar, L. (2026). Bayesian-optimized hierarchical mixture of experts for steel corrosion-rate prediction in cementitious mortars. *Journal of Building Pathology and Rehabilitation, 11*(1), 73. doi:10.1007/s41024-025-00746-x

2 Al Rafi, A., **Momin, Md. F.**, Sen, D., & Chowdhury, A. I. (2026). Effect of parent concrete's properties on compressive strength of recycled aggregate concrete. *Journal of Building Pathology and Rehabilitation, 11*(1), 41. doi:10.1007/s41024-025-00717-2

3 Rahman, T., **Momin, Md. F.**, Podder, S. K., Li, H., & Zheng, P. (2025). Optimizing NGBoost with dynamic sequential model-based optimization for predicting UHPC compressive strength on heterogeneous datasets. *Materials Today Communications*, 112405. doi:10.1016/j.mtcomm.2025.112405

4 Rahman, T., **Momin, Md. F.**, & Provasha, A. A. (2025). Comprehensive analysis of structural parameters influencing the fundamental period of steel-braced RC buildings using machine learning interpretability. *AI in Civil Engineering, 4*(1), 7. doi:10.1007/s43503-025-00051-z

5 Sen, D., Shihab, Md. K., & **Momin, Md. F.** (2025). CNCRP manual based seismic evaluation and retrofitting of existing RC buildings: A case study. *Journal of Rehabilitation in Civil Engineering, 13*(1). doi:10.22075/jrce.2024.32938.1980

6 **Momin, Md. F.**, Islam, Md. W., Masud, F. B., Naomi, N., & Sen, D. (2024). Preliminary and detailed seismic evaluation of an existing RC building in Bangladesh: A case study. *Innovative Infrastructure Solutions, 9*(10), 363. doi:10.1007/s41062-024-01693-1

7 Rahman, T., Sultana, S., Ahmed, T., **Momin, Md. F.**, & Provasha, A. A. (2024). Deep symbolic regression for numerical formulation of fundamental period in concentrically steel-braced RC frames. *Asian Journal of Civil Engineering*. doi:10.1007/s42107-024-01076-y

8 Rahman, T., Hasan, M. H., **Momin, Md. F.**, & Zheng, P. (2023). Data-driven approach to predict the fundamental period of steel-braced RC frames using stacked generalization machine learning models. *Asian Journal of Civil Engineering*. doi:10.1007/s42107-023-00914-9

CONFERENCE PAPER

9 **Momin, Md. F.**, Naomi, N., Masud, F. B., & Sen, D. (2022). Seismic vulnerability assessment of an existing RC building in Bangladesh. *6th International Conference on Civil Engineering for Sustainable Development (ICCESD 2022)*.

TEACHING

Theory courses

- Mechanics of Solids I and II
- Analysis of Statically Determinate Structures
- Design of Steel Structures

Sessional courses

- Design of Concrete Structures Sessional I and II
- Design of Steel Structures Sessional

SKILLS

Engineering software AutoCAD, ETABS, SAP2000, Abaqus

Programming Python, C

Machine learning Ensemble and gradient boosting (NGBoost, stacked generalization), Bayesian and sequential hyperparameter optimization, model interpretability, symbolic regression, validation on heterogeneous datasets

Office tools MS Word, Excel and PowerPoint

Languages Bengali (native), English (fluent)

HONORS AND ACTIVITIES

- Scholarship of Tk. 30,000 from the Pubali Bank Staff Welfare Fund for GPA 5.00 in the SSC examination
- Certificate from Janata Bank Ltd. for brilliant results in the HSC examination
- Planned and organized Civil Job Fiesta 2019, AUST Civil Engineering Society
- General member, AUST Blood Donation Club
- Seminar: "Earthquake Vulnerability Structures in Bangladesh and Design Considerations", AUST

PROFILES

Portfolio Scopus Web of Science Google Scholar ORCID ResearchGate LinkedIn